ARM PrimeCell

STATIC MEMORY CONTROLLER (PL092)

Release Note

PL092 RELN 0000 A

ARM PrimeCell STATIC MEMORY Controller (PL092) Release Note

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The information in this document is Preliminary (information on a product in development).

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- The documents number
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General suggestion for additions and improvements are also welcome.

Feedback on the ARM PrimeCell STATIC MEMORY CONTROLLER

If you have any comments or suggestions about this product, please contact your supplier giving:

- The Product name
- A concise explanation of your comments.

Change history

Issue	Date	Change
A	05 Jun, 2001	First release.

Contents

1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

Table 1.1-1 lists the ARM part numbers for the individual deliverables that form part of the delivery for this ARM PrimeCell peripheral:

Product deliverables for the ARM PrimeCell PL092 peripheral	Document number	Product code
Synthesizable VHDL RTL	N/A	PL092-MN-23001-REL1v0
Synthesizable Verilog RTL	N/A	PL092-MN-22001-REL1v0
Synthesis Scripts (Synopsys Design Compiler, Avant! CB25 cell library)	N/A	PL092-MN-01002-REL1v0
Testbench - Functional Verification, VHDL Testbench - EASY Integration test, VHDL	N/A N/A	PL092-MN-23010-REL1v0 PL092-MN-23012-REL1v0
Testbench - Functional Verification, Verilog	N/A	PL092-MN-22010-REL1v0
Testbench - EASY Integration test, Verilog	N/A	PL092-MN-22012-REL1v0
Test vectors-BusTalk, functional verification	N/A	PL092-VE-01010-REL1v0
Test vectors-BusTalk, integration test	N/A	PL092-VE-01012-REL1v0
ARM PrimeCell STATIC MEMORY CONTROLLER (PL092) Technical Reference Manual (PDF)	ARM DDI 0203 A	PL092-DA-03001-REL1v0
ARM PrimeCell STATIC MEMORY CONTROLLER (PL092) Design Manual (PDF)	PL092 DDES 0000 A	PL092-DC-02002-REL1v0
ARM PrimeCell STATIC MEMORY CONTROLLER (PL092) Integration Manual (PDF)	PL092 INTM 0000 A	PL092-DC-10002-REL1v0
ARM PrimeCell STATIC MEMORY CONTROLLER (PL092) Release Note (PDF)	PL092 RELN 0000 A	PL092-DC-06002-REL1v0

Table 1.1-1: ARM part numbers for ARM PrimeCell SMC PL092

`Note` Only the required deliverables will be supplied, as per the contractual agreement.

For each of the above, two files are generated using the UNIX tar create and the non-document tar files are then UNIX compress'ed. The files are named for each unique product revision, and the generic naming convention is illustrated below :

```
PL092-AA-0196N-REL1v0.tar.Z     - compressed tar file

PL092-AA-0196N-REL1v0.lst       - directory and file listing of above tar file
```

Where `-AA- NNNN` represents the alpha-numeric product sub-type,
and REL1v0 represents the numeric release version of this peripheral.

1.2 File deliverables

This ARM PrimeCell peripheral consists of the following compressed ‘tar’ and ‘list’ file deliverables:

PL092-BU-00000-REL1vN/PL092-BU-01000-REL1v0/PL092-DC-06002-REL1v0.pdf

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/DAtasheet/PL092-DA-03001-REL1v0/PL092-DA-03001-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/DAtasheet/PL092-DA-03001-REL1v0/PL092-DA-03001-REL1v0.tar

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/DoCumentation/PL092-DC-02002-REL1v0/PL092-DC-02002-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/DoCumentation/PL092-DC-02002-REL1v0/PL092-DC-02002-REL1v0.tar

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/DoCumentation/PL092-DC-10002-REL1v0/PL092-DC-10002-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/DoCumentation/PL092-DC-10002-REL1v0/PL092-DC-10002-REL1v0.tar

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-01002-REL1v0/PL092-MN-01002-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-01002-REL1v0/PL092-MN-01002-REL1v0.tar.Z

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-22001-REL1v0/PL092-MN-22001-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-22001-REL1v0/PL092-MN-22001-REL1v0.tar.Z

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-22010-REL1v0/PL092-MN-22010-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-22010-REL1v0/PL092-MN-22010-REL1v0.tar.Z

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-22012-REL1v0/PL092-MN-22012-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-22012-REL1v0/PL092-MN-22012-REL1v0.tar.Z

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-23001-REL1v0/PL092-MN-23001-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-23001-REL1v0/PL092-MN-23001-REL1v0.tar.Z

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-23010-REL1v0/PL092-MN-23010-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-23010-REL1v0/PL092-MN-23010-REL1v0.tar.Z

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-23012-REL1v0/PL092-MN-23012-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/MN_native_model/PL092-MN-23012-REL1v0/PL092-MN-23012-REL1v0.tar.Z

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/VEctors/PL092-VE-01010-REL1v0/PL092-VE-01010-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/VEctors/PL092-VE-01010-REL1v0/PL092-VE-01010-REL1v0.tar.Z

PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/VEctors/PL092-VE-01012-REL1v0/PL092-VE-01012-REL1v0.lst
PL092-BU-00000-REL1v0/PL092-BU-01000-REL1v0/VEctors/PL092-VE-01012-REL1v0/PL092-VE-01012-REL1v0.tar.Z

A generic peripheral user guide document (PDF) is supplied. This describes instructions for synthesis and simulation.
PL092-BU-00003-REL1v0/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.lst
PL092-BU-00003-REL1v0/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.tar

An install script is supplied which uncompresses and tar extracts (un-tar) the compressed tar files in order to build the peripheral directory structure.
PL092-BU-00003-REL1v0/SoftWare/PL000-SW-99000-REL1v0/INSTALL

Note It is possible that the revision number (used in directory paths) for the INSTALL script and user guide may have been updated beyond REL1v0 described above.

A reference cell library (Avant! CB25 0.25µm standard cells) is supplied with this peripheral.
PL092-BU-00003-REL1v0/Avanti/CB25_1999.1.tar.Z
PL092-BU-00003-REL1v0/Avanti/CB25_1999.1.lst

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX compressed tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of compressed tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions cover the following platform/operating system:

- UNIX

2.2 Disk space required

This installation will require 100 Mbytes of free disk space.

2.3 Installation procedure

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the separate products delivered in this shipment.

1. **Relocate the shipment file**

   Relocate the encrypted file (for example <file>.tar.Z.des, if DES encryption has been used) to an appropriate location that has sufficient free disk space.

2. **Decrypt file**

   Follow the accompanying instructions for decryption of the file supplied, to generate a decrypted, compressed, tar file <file>.tar.Z.

3. **Uncompress file**

   Execute the UNIX uncompress command to obtain the tar file <file>.tar:

   ```
   uncompress <file>.tar.Z
   ```

   where <file> is the name of the file supplied.

Note Invoke gzip –d <file>.tar.gz, if gunzip compression has been applied.

4. **Extract tar files**

   To extract individual product tar files, execute the UNIX tar extract command:

   ```
   tar -xvf <file>.tar
   ```

   This will generate individual tar and list files for all of the products within this single shipment. Each peripheral consists of a set of tar files, and each tar file is named with the ARM part number that is listed in the contractual agreement document.

Note It is suggested that part numbers delivered are checked against the contractual agreement document.

2.3.2 Re-creating the peripheral directory structure

The following steps describe how to re-create the directory structure for each separate peripheral, which contains HDL RTL files, synthesis scripts, testbenches and documentation. These steps will need to be repeated for other peripherals in this shipment.

5. Choose method of installation

The directory structure can be extracted using the INSTALL script provided, or manually moving the files to the destination directory and using UNIX uncompress and tar commands. The INSTALL script moves the compressed tar files delivered, to a user specified destination directory and then uncompresses and tar extracts the files to re-create the peripheral directory structure. Manual installation may be preferred if only some of the deliverables are required.

Go to step 6 to use the INSTALL script provided, otherwise go to step 7 for manual installation.

All files begin with PL092, where 080 is a 3-digit value unique to each peripheral.

Note It is important to un-install or rename any version of this peripheral that already exists in the destination directory prior to uncompressing and extracting these tar files.

6. Run the install script

The install script supplied will automatically uncompress and tar extract files for the specified peripheral revision, in a user specified directory. When step 6 is complete there is then no need to perform steps 7 and 8, instead go to step 9.

Change to the directory containing the INSTALL script, for example (note, revision may be later than -REL1v0)

```
cd PL092_PrimeCell/PL092-BU-00003-REL1v0/SoftWare/PL000-SW-99000-REL1v0
```

To invoke the install C-shell script supplied, enter the following at the UNIX command line:

```
INSTALL <PRODUCT_NUMBER> <REVISION> <DESTINATION_DIRECTORY>
```

Options

<PRODUCT_NUMBER> = Each peripheral is defined by unique product code using the characters PL + 3-numeric digits. See below the shipment directory PL092_PrimeCell/

<REVISION> = Each peripheral has a revision number which has a status label and numeric version. See two directories below the shipment directory PL092_PrimeCell/

<DESTINATION_DIRECTORY> = the full directory path where the peripheral is to be installed.

For example

A product shipped with the directory structure /PL092-BU-00000-REL1v0/PL092-BU-00000-REL1v0/ has a product code PL092 and revision number Release REL1v0.

To install this peripheral to the directory /ARM/PrimeCell_peripherals enter the following command line

```
INSTALL PL092 REL1v0 /ARM/PrimeCell_peripherals
```

This will install the peripheral having a product number PL092, with revision number REL1v0, to the destination directory /ARM/PrimeCell_peripherals/

```
Go to step 9
```

7. **Uncompress files**

   Copy all the compressed 'tar' files (*.tar.Z) for the peripheral, to the chosen destination directory. In the example below, the destination directory is called /ARM/PrimeCell_peripherals/.

   Use the UNIX uncompress command to obtain the tar files <file>.tar (* wildcards could also be used):

```
uncompress <file>.tar.Z
```

8. **Extract tar files**

   To extract each file individually, execute the UNIX tar extract command to build the peripheral file structure:

```
tar -xvf <file>.tar
```

9. **Peripheral directory installed**

   When all the peripheral files have been extracted, the destination directory (/ARM/PrimeCell_peripherals/) will contain:

   - the top-level peripheral directory (e.g. SMC_PL092) and underlying file structure of the deliverables, as well as the Release Note document file (PDF).
   - The remaining documentation for the peripheral (in Adobe Portable Document Format, *.pdf), is located in the SMC_PL092/docs/ directory.

2.3.3 Installation of the reference cell library

The reference cell library should be copied manually and tar extracted in the directory containing the PrimeCell peripherals. An example is shown below to copy the Avant! CB25 1999.1 cell library to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PL092_PrimeCell/PL092-BU-00003-REL1v0/Avanti/CB25_1999.1.tar.Z .
```

Execute the UNIX uncompress and tar extract commands as below:

```
uncompress CB25_1999.1.tar.Z
tar -xvf CB25_1999.1.tar
```

2.3.4 Installation of the Generic User Guide

For instructions on performing simulations and synthesis, refer to the generic peripheral user guide.

The most recent revision of the generic user guide should be copied manually to a suitable location. An example is shown below to copy the generic user guide (revision REL1v0) to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PL092_PrimeCell/PL092-BU-00003-REL1v0/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.tar .
```

To extract the document file (PDF), execute the UNIX tar extract command:

```
tar -xvf PL000-DC-08001-REL1v0.tar
```

3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at http://www.adobe.com. Select “Acrobat Reader" and download the reader for your computer platform/operating system.

3.2 Peripheral Documents Supplied

The following documents are supplied for the ARM PrimeCell STATIC MEMORY CONTROLLER PL092:
The documents files are located in the SMC_PL092/docs/ directory.

ARM PrimeCell STATIC MEMORY CONTROLLER PL092
Technical Reference Manual ARM DDI 0203

The technical reference manual provides details for programming and interfacing to this ARM PrimeCell peripheral block.

ARM PrimeCell STATIC MEMORY CONTROLLER PL092
Integration Manual PL092 INTM 0000

The integration manual provides information required to integrate this ARM PrimeCell block within a typical industry standard ASIC design flow.

ARM PrimeCell STATIC MEMORY CONTROLLER PL092
Design Manual PL092 DDES 0000

The design manual describes the implementation details of this ARM PrimeCell block and the testbenches and test programs used for functional verification.

4 TRIAL TESTING

4.1 Product Maturity

This product is newly developed IP and approved for final release.
At the time of release REL1v0, it has not been implemented in production silicon.

4.2 Technical data summary

The following is a summary of the key technical points from testing of the ARM PrimeCell STATIC MEMORY CONTROLLER PL092.

	VHDL RTL	Verilog RTL
Cell library	Avant! Passport 1999.1 CB25 v2.1 (0.25 µm)	Avant! Passport 1999.1 CB25 v2.1 (0.25 µm)
Total cell area, without scan, NAND-2 equivalents	9,067	9,179
Net interconnect area, without scan, NAND-2 equivalents	4,687	4,701
Total cell area, scan inserted,NAND-2 equivalents	10,954	11,124
Net interconnect area, scan inserted,NAND-2 equivalents	5,870	5,907
Clock frequency used	100MHz	100MHz
Scan insertion figure % (number of ‘D’ flip-flops replaced)	100% (678)	100% (1621)
Estimated scan fault coverage, collapsed.	99.80% (+-2%)	99.80% (+-2%)
Synthesis and scan test tools and version	Synopsys Design Compiler and Test Compiler 2000.11	Synopsys Design Compiler and Test Compiler 2000.11
Simulation tools and versions	Model Technology Inc, ModelSim v5.4e	Model Technology Inc, ModelSim v5.4e Cadence LDV 3.1
Global Peripheral Development Environment Version	Release REL1v4	

4.3 BusTalk Functional block verification test scenarios

Functional block verification has been performed with the following scenarios using the BusTalk testbench with free-running HCLK.

Note Simulation log files are only supplied for RTL simulations for this beta release.

4.3.1 Synchronous clocks

HCLK	Simulations
100MHz (10ns period)	VHDL and Verilog, RTL and gate-level min/max SDF

For further details of functional verification testing, refer to the ARM PrimeCell STATIC MEMORY CONTROLLER (PL092) Design Manual.

5 DIFFERENCES FROM PREVIOUS REVISIONS

6 KNOWN ISSUES

No known issues.

7 REVISION HISTORY

Date	Revision	Description
05-Jun-2001	REL1v0	Initial Release 1.0